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Fractal Neural Networks for Radio-Frequency Interference Detection

Abstract

This proposal outlines a study to design, implement, and evaluate compact, self-similar Fractal Neural Networks (FNNs) for the automated detection and restoration of radio-frequency interference (RFI) in radio-astronomy time–frequency data. RFI degrades sensitivity and increases false positives across instruments such as the Murchison Widefield Array (MWA), LOFAR, and MeerKAT. Classical rule-based flagging tools remain strong operational baselines but may require per-dataset tuning and often do not quantify downstream scientific effects. Recent deep learning (DL) approaches show promise for segmentation and inpainting, yet reproducibility varies and science-level impacts are not consistently reported. This study will develop lightweight fractal/Fibonacci-inspired variants optimized for spectrograms, benchmark them against established baselines, and quantify scientific impact through transient recovery and power-spectrum bias analyses. Methods include PyTorch implementations, AOFlagger-labelled real data, controlled synthetic RFI injections, and rigorous evaluation balancing pixel-level and science-level metrics. The expected contribution is a reproducible, efficiency-aware RFI pipeline that improves data quality and operational practicality, with openly documented artefacts for community reuse.

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1. Introduction

Radio-frequency interference (RFI) remains a central obstacle to sensitivity and reliability in radio astronomy. It contaminates time-frequency spectrograms, reduces transient detectability, and biases power-spectrum inferences in survey pipelines. As data rates and real-time demands grow, robust and efficient RFI handling becomes essential for scientific yield and sustainable operations.

1.1. Research aims and objectives

Aim: To determine whether compact, self-similar FNNs can match or exceed established RFI detection/restoration baselines while improving efficiency and deployability.

Objectives:

1. Design and implement a family of fractal/Fibonacci-scaled FNNs tailored to spectrogram segmentation and inpainting.
2. Benchmark against AOFlagger, U-Net, and efficient CNN/ViT baselines on real and synthetic datasets.
3. Quantify downstream scientific impact via transient injection/recovery and power-spectrum bias assessments.
4. Package reproducible code, configurations, and containers for community adoption.

1.2. Background and problem statement

Classical morphology-aware masking, notably AOFlagger, is widely adopted in operations due to its robustness and transparency (Offringa et al., 2010, 2015, 2023). However, optimal parameterisation can be dataset‑dependent, and its scientific impact is typically assessed indirectly. DL methods have reported strong segmentation performance, yet comparisons are difficult to reproduce, and efficiency constraints for observatory use are not always addressed (Akeret et al., 2017; Vafaei Sadr et al., 2020; Yang et al., 2020). The problem addressed here is the lack of compact, rigorously‑evaluated architectures that balance accuracy with resource efficiency and that report science‑level consequences alongside pixel‑level scores.

1.3. Purpose and theoretical framework

The study will evaluate self‑similar (fractal) architectures that provide multi‑scale receptive fields and controlled parameter growth (Larsson et al., 2016), combined with efficiency‑oriented design ideas (Howard et al., 2017; Tan & Le, 2019). The framework integrates standard signal‑processing views of spectrogram structure with modern DL segmentation/restoration paradigms to support end‑to‑end scientific integrity.

1.4. Significance

If successful, the work will yield models that are both accurate and practical to deploy, with transparent, reproducible evaluations. This directly benefits facilities facing heterogeneous RFI environments and limited compute, and it contributes evidence on how masking/inpainting choices propagate to science products.

1.5. Feasibility, scope, and limitations

The study will leverage public archives (subject to licences), containerised AOFlagger, PyTorch implementations, and synthetic injection for controlled ground truth. Feasibility is enhanced by compact models and CPU-first defaults, with optional GPU acceleration. Limitations include dataset heterogeneity, imperfect labels, and constrained access to large accelerators. These risks will be mitigated via human-validated subsets, label-noise-aware training where appropriate, cross-dataset tests, and careful resource profiling.

2. Preliminary Literature Review

Work on RFI treatment in radio astronomy spans mature, rule-based flagging systems and a growing body of ML research. The AOFlagger family remains the de facto classical tool: its SumThreshold and morphology-aware operations are widely used to produce quality masks and to support pipeline automation (Offringa et al., 2010; Offringa et al., 2015; Offringa et al., 2023). AOFlagger's strength is robustness across many observing conditions and its acceptance in operational pipelines; its principal limitations are the need for per-dataset tuning in some cases and the fact that algorithmic heuristics do not by themselves quantify how masking choices propagate to downstream science metrics.

In parallel, DL methods applied to dynamic spectra have shown clear promise for both detection and restorative tasks. Convolutional segmentation networks and U-Net variants were among the first widely-published ML approaches for RFI masking (Akeret et al., 2017), and subsequent work has demonstrated that residual and transfer-learning based

classifiers (e.g. ResNet-derived models) achieve strong detection accuracy on several telescope datasets (Vafaei Sadr et al., 2020; Yang et al., 2020). More recent architectures introduce multi-scale attention and efficient building blocks to improve segmentation quality while reducing compute (Gu et al., 2024). Complementary work reframes RFI treatment as restoration or inpainting, replacing masked samples rather than only predicting masks, and evaluates the fidelity of restored spectra for specific science use-cases such as pulsar timing (Zhang et al., 2024) and Epoch-of-Reionisation analyses (Pagano et al., 2023).

From an architectural and systems perspective there are two relevant strands. First, compact and efficient model design (MobileNet, EfficientNet concepts) offers practical trade-offs between accuracy and latency that are essential for adoption in observatory contexts (Howard et al., 2017; Tan & Le, 2019). Second, self-similar or fractal macro-architectures (FractalNet) provide an alternative route to multi-scale receptive fields without simple depth scaling and have attractive properties for subnetwork extraction and parameter control (Larsson et al., 2016). On the systems side, efforts to bring AOFlagger-style routines into high-throughput or real-time environments underscore the engineering constraints that any ML solution must respect (Sclocco et al., 2020; van Nieuwpoort, 2017).

Strengths across the field include robust classical baselines, demonstrable ML accuracy gains on curated datasets, and emerging restoration approaches that consider scientific fidelity. Weaknesses and challenges are nonetheless clear: many ML studies assume

plentiful compute and focus on per-pixel accuracy without fully quantifying downstream scientific impact; comparisons are often limited by inconsistent labeling and by the practical difficulty of reproducing AOFlagger configurations; and there is relatively little exploration of compact, self-similar architectures specifically applied to spectrogram segmentation and restoration.

These observations justify the proposed research. The combination of fractal/Fibonacci scaling with modern, efficiency-aware convolutional blocks targets the gap between high-performance ML models and models that are resource-appropriate and amenable to operational deployment. Equally important is rigorous evaluation: controlled synthetic injections, AOFlagger-derived baselines and small human-validated subsets together provide a practical, reproducible framework for measuring both detection/restoration accuracy and downstream scientific consequences (for example transient recovery and power-spectrum bias). By situating fractal architectures within this broader evaluation strategy, the project addresses a concrete and under-explored gap, producing compact, well-characterised models and reproducible evidence about their practical utility.

3. Experimental Design and Methods

3.1. Design overview

The study will evaluate RFI detection and restoration on two-dimensional spectrograms. The pipeline will combine:

1. Real data labelled by containerised AOFlagger (to preserve exact baseline behaviour).

2. Synthetic injections spanning RFI morphologies, duty cycles, and SNRs to provide controlled ground truth for restoration and stress testing.
3. Small human-validated subsets to assess label quality and calibrate label-noise models.

3.2. Sequence of activities

1. Data acquisition and baselining: ingest public archives (e.g., MWA, LOFAR, MeerKAT, FAST, Apertif where licences permit); generate AOFlagger masks inside Docker; document ingestion and configuration.
2. Model development: implement a compact FNN family (fractal/Fibonacci scaling) in PyTorch; implement baselines (U-Net, ResNet-derived, efficiency-aware CNN blocks, and a compact ViT where informative).
3. Training and ablations: run deterministic experiments across seeds and splits; conduct ablations on scaling, regularisation (e.g., drop-path), separable convolutions, and channel schedules.
4. Restoration/inpainting: apply learned inpainting where appropriate; measure fidelity.
5. Downstream science tests: perform transient injection/recovery (TPR/FPR) and estimate power-spectrum bias/variance with established EoR procedures.
6. Packaging and reproducibility: archive configs, checkpoints, and container recipes; provide clear documentation.

3.3. Analysis plan

- Segmentation metrics: Intersection over Union (IoU), Dice, precision, recall, F1, and precision–recall curves.
- Restoration metrics: PSNR, SSIM, and analysis of structured residuals across time and frequency.
- Science-level metrics: transient detection TPR/FPR on controlled injections; power-spectrum bias and variance estimations.
- Statistical treatment: report mean ±95% CIs across seeds/splits; use paired tests where applicable; track latency, peak memory, parameter counts, and FLOPs to quantify efficiency.

3.4. Tools and justification

- PyTorch for modelling/profiling and deterministic runs.
- AOFlagger as operational baseline, containerised to ensure reproducibility.
- Synthetic injection to provide controlled, diverse ground truth beyond noisy labels.
- CPU-first defaults to ensure accessibility; GPU acceleration when available.

3.5. Risks and mitigations

- Heterogeneous data/labels: use documented ingestion, cross-dataset tests, and small human-validated subsets.
- Compute constraints: prioritise compact designs; use proxy datasets and early-stopping; profile and report efficiency metrics.

- Reproducibility: fixed seeds, archived configs, exact container images, and public artefacts.

4. Proposed timetable

Weeks	Activities (summary)	Key deliverables
1–3	Finalise proposal; complete focused literature review and gap analysis; inventory public datasets and access terms; containerise and verify AOFlagger; prototype synthetic RFI injection; smoke-test experiment scaffolding.	Annotated literature summary; dataset inventory; confirmed AOFlagger-in-Docker; injector prototype; seeded repo with configs.
4–7	Curate and pre-process spectrograms; run AOFlagger to produce baseline masks; generate synthetic datasets (morphology, SNR, duty cycle grid); create human-validated subsets; version all splits.	Cleaned datasets; baseline masks; synthetic dataset; validation subset; versioned splits.
6–9	Implement FNN family and baselines (U-Net, ResNet-derived, efficient CNN blocks, compact ViT if used); add	Model code; training scripts; example checkpoints; profiling reports.

	deterministic training, checkpointing, logging, profiling.	
9–12	Train and run ablations (scaling, regularisation, separable convs); measure performance and resource metrics across seeds/splits.	Ablation results; latency/memory/params/FLOPs tables; shortlist candidates.
13–15	Run restoration/inpainting and downstream tests (transient recovery; power-spectrum bias/variance); evaluate cross-dataset generalisation and label-noise sensitivity.	Restoration metrics; downstream plots/tables; robustness analysis.
16–20 (optional)	Reproducibility runs; deployment optimisation for top models; package containers/configs/checkpoints; prepare final report figures and documentation.	Reproducible artefacts; deployment notes; final write-up.

Table 1

Proposed timetable

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